



COMPREHENSIVE PATENT SEARCH REPORT

April 10, 2025

Invention: Quantum Key Distribution (QKD) System Utilizing Super-positional Entropy for Post-Quantum Cryptography
Case Number: TPS89575

Kircher, Jed:

A patent search has been conducted and a report prepared. A patent search report discovers inventions similar to yours. Each similar invention found is called “prior art” and are all inventions disclosed before yours. To be granted a patent by the US Patent & Trademark Office (USPTO), a Patent Office examiner must determine that your invention meets the three patenting criteria of being novel, non-obvious, and useful when compared to prior art.

Having a patent search report provides you with valuable information. The report will show you prior art that may be cited by the USPTO when they look at a patent application for your invention. Using this information, you can make a preliminary assessment towards whether or not the USPTO will consider your patentable. You can then decide if you wish to continue to file a patent application. In addition, you can make changes or improvements to your invention to make your invention more unique and therefore more patentable, before submitting a patent application.

Upon reviewing this search report, determine whether you can make improvements to your invention to increase the uniqueness of your invention above the prior art. Send any such improvements to us before deciding to proceed with a patent application. If you have any questions and wish to discuss the search report, please let us know. We will wait for your decision to proceed with a patent application.

Sam Yuan
U.S. Licensed Patent Practitioner
Thoughts to Paper
20251 Century Blvd Ste 140
Germantown, MD 20874

NO GUARANTEE OF FINDING ALL SIMILAR INVENTIONS OR FUTURE PATENT APPROVAL. NOT AN INFRINGEMENT OPINION OR FREEDOM TO OPERATE OPINION

You understand and accept that there is absolutely no guarantee that we will find all similar inventions (prior art), or that we will find all prior art that may be used by a patent examiner to potentially reject your patent application should you decide to file a patent application in the future. Whether a prior art is considered similar enough to your invention is a subjective determination which will vary from individual to individual. Therefore, it is possible that we may find prior art that a patent examiner will not use in a patent application rejection or conversely, a patent examiner may use a prior art in a patent application rejection that we do not find. Patent searching is not an exact science and it is common for prior art to not be found. Further, recently filed patent applications not published cannot be found in a patent search. A patent search should not be relied upon as a definitive conclusion as to whether your invention can or cannot obtain a patent. Instead, a patent search should be used as a general assessment of the patentability of your invention, as well as for use in improving your invention over existing inventions to increase your overall patentability. Any patentability opinion that is provided in the search report is a professional opinion only and comes with absolutely no guarantee of the outcome of a patent application. You agree to hold us harmless for any actions or inactions you may take while relying on the search report we provide. Should a patent application be rejected, there is no refund of any fees paid, including this patent search fee. You agree not to seek losses from us for any costs you may have incurred through reliance of our patent search.

A patent search report gives a general assessment as to the likelihood an invention can obtain a patent. It does not give an opinion as to you have the freedom to sell your invention without infringing upon another's patent. It is possible for your invention to be patentable and obtain a patent, and yet you cannot make the invention without licensing parts of the invention from another patent holder. A patent search discusses patentability only, the likelihood you can obtain a patent.

TABLE OF CONTENTS

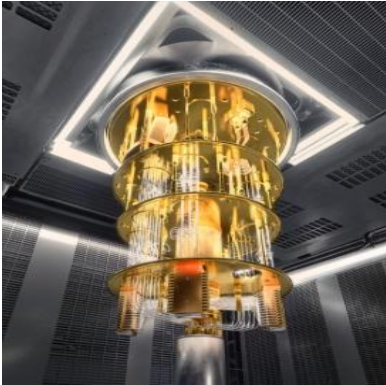
PRIOR ART	4
UNIQUE ASPECTS OF YOUR INVENTION.....	8
UTILITY AND/OR DESIGN SUITABILITY.....	8
CHANCES OF APPROVAL FOR UTILITY PATENT	9
CHANCES OF APPROVAL FOR DESIGN PATENT	9
ANALYSIS & RECOMMENDATIONS	10
RECOMMENDED STEPS FOR UTILITY PATENT.....	12
PROVISIONAL OR NON-PROVISIONAL? HOW TO DECIDE	13
RECOMMENDED STEPS FOR DESIGN PATENT	14

PRIOR ART

Prior art which is relevant to your invention is listed in the table below. For each prior art, a rank number is assigned which represents how important the particular prior art is to your invention. A smaller rank number represents a prior art that is more relevant than a prior art with a larger rank number. Full copies of the prior art listed are in the Prior Art Portfolio, a separate file or document.

When browsing through the prior art, it will be beneficial to write down how your invention is different and how it may overcome the prior art's shortcomings. Including this information will benefit your patent application.

Rank	Document Number	Invention Title / Similarities Noted & Comments
4	Online prior art	Advancements in Secure Quantum Communication and QKD The prior art includes a disclosure of advancements in quantum communication and QKD protocols like BB84 and E91 that rely on quantum physics principles, such as the no-cloning theorem and uncertainty principle, for secure key exchange. The prior art aligns by utilizing quantum superposition principles to establish post-quantum secure cryptographic keys and detecting eavesdropping through quantum states. Unlike the prior art, your invention includes a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism. https://tinyurl.com/2h2cjypr *Please note that non-patent references can be used as prior art. This particular prior art was found in the web*

4	Online prior art	Quantum Hacking and Quantum System Vulnerabilities The prior art includes a disclosure of attack vectors and real-world security risks in QKD, with emphasis on how current technologies resist interception and maintain secure communication. Unlike the prior art, your invention includes a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism. https://tinyurl.com/bdfjefn2  *Please note that non-patent references can be used as prior art. This particular prior art was found in the web*
---	------------------	--

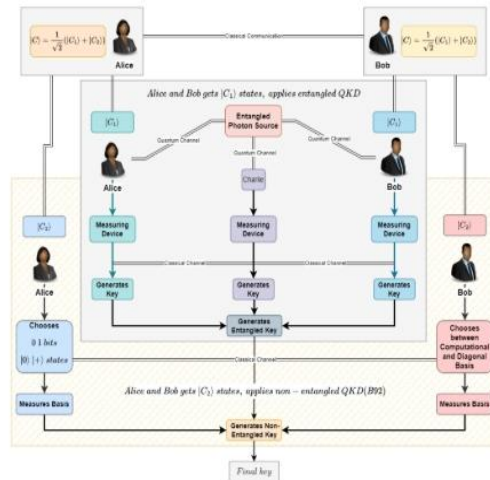
5

Online prior art

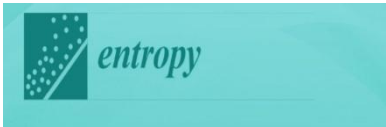
Combining Entangled and Non-Entangled based Quantum Key Distribution Protocol with GHZ state

The prior art includes a disclosure of a hybrid QKD protocol combining entanglement and superposition to probabilistically boost security and efficiency. Unlike the prior art, your invention includes a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism.

<https://arxiv.org/html/2411.06586v1>



Please note that non-patent references can be used as prior art. This particular prior art was found in the web

5	Online prior art	Physical-Layer Security in QKD The prior art includes a disclosure of physical-layer security principles in QKD systems, emphasizing enhancements like secret key distillation and secure key synchronization under quantum physics constraints. Like your invention's entropy-validation mechanisms, this study underlines the importance of secure key verification during distribution. Unlike the prior art, your invention includes a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism. https://tinyurl.com/3cv2bjmx  *Please note that non-patent references can be used as prior art. This particular prior art was found in the web*
5	Online prior art	Advances in Chip-Based Quantum Key Distribution The prior art includes a disclosure of integrated QKD systems leveraging photon sources, detectors, and advanced encoding to improve security and scalability. Unlike the prior art, your invention includes a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism. https://tinyurl.com/f5267msk  *Please note that non-patent references can be used as prior art. This particular prior art was found in the web*

UNIQUE ASPECTS OF YOUR INVENTION

These are components or aspects of your invention which are unique when compared to the prior art. These will be used to argue approval of a patent application. Ensure that unique aspects of your invention are clearly explained in detail. It is beneficial to have as many unique aspects as possible. You may be able to improve on the similarities in the table at the top of the page and convert them into a unique aspect for your invention.

A specific combination of ALL aspects used in your invention:

- a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism.
- The specific appearance of your invention (design patentability merit)

UTILITY AND/OR DESIGN SUITABILITY

Utility patents and design patents work in parallel with each other. One is not a replacement for the other. Utility patent protects functionality whereas design patent protects only the exterior shape of a product. For example, Apple has utility patents to protect the iPhone's function but also has design patents to protect the iPhone's exterior body shape.

Below is whether your invention is suitable for utility patent and design patent. If suitable for both, you can decide to file both to protect the function (utility) and exterior shape (design).

SUITABLE FOR UTILITY PATENT

Protects the functionality, components, and how it works.

Yes

SUITABLE FOR DESIGN PATENT

Protects the exterior shape. Does not protect functionality.

**Yes;
If provided the FINAL appearance of your invention's shape.**

CHANCES OF APPROVAL FOR UTILITY PATENT

Utility Patent Protects: The invention's functionality, components, and how it works.

Our opinion as to the likelihood your invention will pass each of the three patentability criteria if submitted to the US Patent Office as a non-provisional utility patent application.

NOVEL

Fair

NON-OBVIOUS

Low

USEFUL

Good

OVERALL CHANCE OF UTILITY APPROVAL

Fair;

Your invention has an overall fair chance of patentability compared to the prior art findings above (see analysis and recommendations).

CHANCES OF APPROVAL FOR DESIGN PATENT

Design Patent Protects: The invention's exterior shape only, with no protection of functionality.

Based on the patent search and our professional experience, below is the likelihood your invention will pass each of the three patentability criteria if submitted to the US Patent Office as a design patent application.

NOVEL

Good

NON-OBVIOUS

Good

USEFUL

Good

OVERALL CHANCE OF DESIGN APPROVAL

Good

ANALYSIS & RECOMMENDATIONS

Our analysis as to the patentability of your invention along with any recommendations to increase the chances of patent application approval.

A thorough search was conducted, and relatively few relevant patents were found. To identify the novelty of your invention, we would determine whether a single prior art reference disclosed all of your invention features. The absence of such a reference would indicate that you have a good chance of the patent examiner considering your invention to be novel.

Based on my review of the patents or online disclosure generated in this report, it appears that the concept of your invention may be new, constituting a fair merit of novelty. Online prior art **[Advancements in Secure Quantum Communication and QKD]** is identified as the most relevant document. The prior art includes a disclosure of advancements in quantum communication and QKD protocols like BB84 and E91 that rely on quantum physics principles, such as the no-cloning theorem and uncertainty principle, for secure key exchange. The prior art aligns by utilizing quantum superposition principles to establish post-quantum secure cryptographic keys and detecting eavesdropping through quantum states. Unlike the prior art, your invention includes a Quantum Key Distribution (QKD) system that securely generates, validates, and exchanges cryptographic keys using quantum superposition entropy, ensuring post-quantum security through intrinsic randomness, eavesdropping detection, and a self-correcting validation mechanism.

Although the novelty of your invention appears to be fair, an important point to note is that three requirements must be met for patentability: novelty, non-obviousness, and usefulness. Your invention may easily satisfy the last of these because it is not a perpetual motion machine; therefore, the patent examiner would readily consider it useful. However, the question then becomes whether the differences between your invention and the prior art are such that your invention as a whole would have been obvious to the one knowledgeable in the technical field. Your invention would be obvious, for example, if it is considered to be already known in view of several references in combination.

Your invention must be sufficiently different from what has been used or described before so that it may be said to be non-obvious to a person with ordinary skill in the area of technology related to the invention. For example, the substitution of one color, material, or quantity for another, or changes in size, are ordinarily not patentable because they are considered obvious factors in an invention.

The prior arts provided above may lead to relevant information on the issue of obviousness. For example, the Online prior art **[Advancements in Secure Quantum Communication and QKD]** does not include all the features in your invention; however, a modification to include such a feature can be considered obvious by a patent examiner in view of the prior arts.

Based on the prior art, your chances of meeting the non-obvious criteria are low, because, as discussed above, a patent examiner can use a specific combination of relevant prior art findings to reject your patent application based on your invention being considered obvious.

Overall, these prior art findings should serve as references for relevant information to make your invention even more unique and non-obvious. We strongly recommend that you focus on highlighting aspects of your invention that make it unique. The more components you can add and elaborate in your invention, the greater the chances you can get the patent. Additional components that are distinguished from the prior art findings can be added.

Let us know you are ready to proceed with a patent application once you have concluded that you have maximized the novelty and non-obviousness of your invention in light of the prior art on the concepts mentioned earlier.

You can choose to proceed with a provisional patent application, which, when filed, gives you 12 months of "patent pending" status. You also have the option to proceed directly with a full utility patent application, which is subjected to examination and, if approved, will grant you 20 years of full patent protection rights on the utility of your invention. If you think the overall look, shape, or form is unique and important to your invention, you may want to consider pursuing a design patent. If approved, this will grant you 15 years of full patent protection rights on the external appearance of your invention. The design patent can protect the specific shape of your invention and will most likely be approved if views of the invention are consistent. Very rarely is prior art found that exactly conforms to the ornamental design of an invention. Some inventors choose to proceed with both a utility patent and a design patent to protect both the function and the design.

Should you decide to move forward with a provisional patent application, I strongly recommend that you select our essentials-level or professional-level package, which includes professional writing of your patent application

details. Our engineers will be able to fully elaborate on the unique aspects of your invention in light of the prior art to maximize the value of your provisional patent application. Contact your consultant to discuss the search results in more detail or to proceed with the patent process.

RECOMMENDED STEPS FOR UTILITY PATENT

If you decide to file a utility patent application,

Decide your path

PROVISIONAL UTILITY PATENT APPLICATION

12 months temporary. Option to convert to a non-provisional later, within 12 months.

Provisional Patent Application Basic Package

Provisional Patent Application Essentials Package:

+ Professional Technical Writing **OR**
+ Professional Patent Drawings

Provisional Patent Application Professional Package:

+ Professional Technical Writing **AND**
+ Professional Patent Drawings

Within 12 months

Non-Provisional Patent Application

All parts required by the Patent Office,
including technical writing, patent
drawings, patent claims.

NON-PROVISIONAL UTILITY PATENT APPLICATION

20 years of patent rights, if approved

Non-Provisional Patent Application

All parts required by the Patent Office,
including technical writing, patent
drawings, patent claims.

Recommended
for you

Recommended
for you

RECOMMENDATION NOTES

Your current disclosure of your invention has an overall rating of **fair**, where your invention may be considered as a fairly novel and your chances of meeting the non-obvious criteria are low. We highly recommend further additions/improvements to make your invention sufficiently different from what has been used or described before that it may be said to be non-obvious to a person with ordinary skill in the area of technology related to the invention. Once confident and ready to proceed, let your invention consultant know you want to proceed with either the provisional or non-provisional patent application using the guidelines above. We strongly recommend a professional package level provisional patent application because it will include a conference call with our engineers to flesh out your unique ideas and detail them. We will also prepare several pages of technical descriptions and professional patent drawings to add value to your provisional patent application. Any further pictures, specifications, and disclosures of updated prototypes (if available) will be necessary for the patent application. We need to thoroughly elaborate and emphasize the unique aspects of your invention in the provisional patent application to ensure these unique aspects are properly protected.

PROVISIONAL OR NON-PROVISIONAL? HOW TO DECIDE

Decide your path

PROVISIONAL UTILITY PATENT APPLICATION

12 months temporary. Option to convert to a non-provisional later, within 12 months.

COMMON SITUATIONS:

- 1) You are not ready to financially commit to the higher cost associated with a non-provisional and you want 12 months to make the decision or to gather funds.
- 2) There is a high chance you will make major changes to your invention within the next 12 months.

NON-PROVISIONAL UTILITY PATENT APPLICATION

20 years of patent rights, if approved

COMMON SITUATIONS:

- 1) You are ready to financially commit to the higher cost associated with a non-provisional patent application.
- 2) Your invention is generally finalized and there is a low chance of major changes.

Use the provisional patent application if you need 12 months of time to decide if you will commit to a non-provisional patent application. You can use the 12 months to raise funds, test out the market, get feedback, improve upon the invention, or any other way to *gain confidence* before committing the additional costs to a non-provisional patent application.

However, if you don't need 12 months of time to make the decision to file a non-provisional, you can skip the provisional application and go straight for the non-provisional. Generally, you go directly to the non-provisional application if you meet two conditions. First, you are financially ready and willing to commit the larger investment, as the non-provisional application is always more expensive than the provisional. Second, you do not anticipate making changes to your invention. Once a patent application is filed, it cannot be changed. If you make changes to your invention, you would have to file a new patent application. Most inventors want to make sure that when they do invest in filing a non-provisional patent application, the invention is generally finalized. If you make major changes to the invention after filing a non-provisional, you would have to file a second non-provisional patent application to include those new changes.

RECOMMENDED STEPS FOR DESIGN PATENT

Remember, a design patent application works in parallel with a utility patent application. Design patents offer additional protection for the exterior shape of your invention only. It does not protect functionality.

Because your invention is also suitable for a design patent, decide if you want to also file a design patent to cover the exterior shape of your invention. If so, let us know.

Remember, a design patent does not protect any components, functionality, or how an invention works. It only protects the exterior shape. The design patent is best used one of two ways:

- 1) **In conjunction with a utility patent.** You file both utility patent and design patent to protect both function and design.
- 2) **If you do not care about the function.** If you do not care about protecting what your invention does, but only want to protect the exterior shape, you can file just a design patent. This is common for when someone already has a utility patent on your invention. For example, if you made a modern looking chair, you are not the first to invent the chair but have instead invented the unique exterior design only.